Christmas Party

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1717 | Mathematics | https://cses.fi/problemset/task/1717

There are n children at a Christmas party, and each of them has brought a gift. The idea is that everybody will get a gift brought by someone else.

In how many ways can the gifts be distributed?

Input

The only input line has an integer n: the number of children.

Output

Print the number of ways modulo 10^9+7.

Constraints

- $1 \le n \le 10^6$

Example

Input
4

Output
9